

The screenshot displays the Verilator GUI interface, which is used for simulating digital circuits. The top bar shows the current file being viewed, "audio_soc", and includes navigation controls like back, forward, and search. Below the top bar, there's a tabbed view of open files, with "3.waveform_AXI_Peripheral_V1.1.vcd" currently selected.

The main window is divided into three sections:

- Filter tree:** A sidebar on the left containing a hierarchical list of signals. It starts with "TOP" and lists various internal components and their outputs, such as "TOP.audio_out", "TOP.audio_soc_top.clk", and "TOP.audio_soc_top.rst_n".
- VCD View:** The central area displaying the waveform data. It shows a grid where each row corresponds to a signal from the filter tree. The signals are plotted against time, with values represented by red horizontal bars. Some signals show constant values (e.g., "x" for unknown or "0xx" for zero), while others show transitions over time.
- Cursor:** A vertical line indicating the current position in the simulation timeline, labeled with time values like "00", "500.00", "1000.00", etc.

At the bottom of the screen, there's a status bar showing system information, including the number of errors/warnings (86) and the current port configuration (Port : 5500).

audio_soc

Update

.gitignore

alu.sv

3.waveform_AXI_Peripheral_V1.1.vcd

index.html

styles.css

app.js

audio_soc_top.sv M

audio_pwm_axi.sv

stream.py 6

server.js 9+

3.waveform_AXI_Peripheral_V1.1.vcd

VCD

rm *

Zoom

Cursor

HOME

END

Filter tree

TOP

TOP.audio_soc_top.u_audio.s_axi_awready

x

TOP.audio_soc_top.u_audio.s_axi_wdata [31:0]

0xx

0xx

TOP.audio_soc_top.u_audio.s_axi_wvalid

0

TOP.audio_soc_top.u_audio.s_axi_wready

1

TOP.audio_soc_top.u_audio.s_axi_bresp [1:0]

0x0

0x0

TOP.audio_soc_top.u_audio.s_axi_bvalid

0

TOP.audio_soc_top.u_audio.s_axi_bready

1

TOP.audio_soc_top.u_audio.audio_out

0

TOP.audio_soc_top.u_audio.sample_reg [7:0]

0x00

0x00

TOP.audio_soc_top.u_audio.counter [7:0]

0x00

TOP.audio_soc_top.u_core.clk

x

TOP.audio_soc_top.u_core.rst_n

x

TOP.audio_soc_top.u_core.mem_addr [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.mem_wdata [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.mem_we

0

TOP.audio_soc_top.u_core.mem_rdata [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.pc_curr [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.pc_next [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.instr [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.reg_d1 [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.reg_d2 [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.alu_res [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.imm [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.alu_mux_out [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.wb_data [31:0]

0xx

0xx

TOP.audio_soc_top.u_core.reg_we

x

TOP.audio_soc_top.u_core.alu_s

x

TOP.audio_soc_top.u_core.mem_reg

x

TOP.audio_soc_top.u_core.alu_op [1:0]

0xx

0xx

TOP.audio_soc_top.u_core.alu_ctrl [3:0]

0xx

0xx

TOP.audio_soc_top.u_core.func3 [2:0]

0x0

0x0

TOP.audio_soc_top.u_core.func7 [6:0]

0x20

0x0

00

500.00

1000.00

1500.00

2000.00

2500.00

3000.00

3500.00

4000.00

4500.00

5000.00

Credits: Benedek Racz: [GitHub](#)

main* 86 6 AWS

Port : 5500

[illegible]

audio_soc

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VCD

rm *

Zoom

HOME

<

>

END

Cursor

Filter tree

TOP

TOP.audio_soc_top.u_core.u_regfile.rf[17] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[18] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[19] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[20] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[21] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[22] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[23] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[24] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[25] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[26] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[27] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[28] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[29] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[30] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_core.u_regfile.rf[31] [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_imem.addr [31:0]

0xx

0xx

TOP.audio_soc_top.u_imem.instr [31:0]

0x40000293

0x40000293

TOP.audio_soc_top.u_pc.clk

x

TOP.audio_soc_top.u_pc.rst_n

x

TOP.audio_soc_top.u_pc.irq

x

TOP.audio_soc_top.u_pc.d [31:0]

0x00000004

0x00000004

TOP.audio_soc_top.u_pc.q [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_pc.mepc [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_timer.clk

1

TOP.audio_soc_top.u_timer.rst_n

0

TOP.audio_soc_top.u_timer.we_i

0

TOP.audio_soc_top.u_timer.wdata_i [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_timer.irq

0

TOP.audio_soc_top.u_timer.count [31:0]

0x00000000

0x00000000

TOP.audio_soc_top.u_timer.threshold_reg [31:0]

0x00000000

0x00000000

00

50.00

100.00

150.00

Credits: Benedek Racz: [GitHub](#)

main* 86 6 AWS

Port : 5500

audio_soc

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3.waveform_AXI_Peripheral_V1.1.vcd

VCD

rm *

Zoom

Cursor

HOME

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→

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>

↶

↷

END

Filter tree

TOP

TOP.audio_soc_top.u_core.mem_reg

TOP.audio_soc_top.u_core.alu_op [1:0]

TOP.audio_soc_top.u_core.alu_ctrl [3:0]

TOP.audio_soc_top.u_core.func3 [2:0]

TOP.audio_soc_top.u_core.func7 [6:0]

TOP.audio_soc_top.u_core.u_alu.a [31:0]

TOP.audio_soc_top.u_core.u_alu.b [31:0]

TOP.audio_soc_top.u_core.u_alu.alu_ctrl [3:0]

TOP.audio_soc_top.u_core.u_alu.result [31:0]

TOP.audio_soc_top.u_core.u_alu.zero

TOP.audio_soc_top.u_core.u_ctrl.opcode [6:0]

TOP.audio_soc_top.u_core.u_ctrl.reg_write

TOP.audio_soc_top.u_core.u_ctrl.alu_src

TOP.audio_soc_top.u_core.u_ctrl.mem_to_reg

TOP.audio_soc_top.u_core.u_ctrl.alu_op [1:0]

TOP.audio_soc_top.u_core.u_regfile.clk

TOP.audio_soc_top.u_core.u_regfile.we

TOP.audio_soc_top.u_core.u_regfile.rs1 [4:0]

TOP.audio_soc_top.u_core.u_regfile.rs2 [4:0]

TOP.audio_soc_top.u_core.u_regfile.rd [4:0]

TOP.audio_soc_top.u_core.u_regfile.wd [31:0]

TOP.audio_soc_top.u_core.u_regfile.rd1 [31:0]

TOP.audio_soc_top.u_core.u_regfile.rd2 [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[1] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[2] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[3] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[4] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[5] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[6] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[7] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[8] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[9] [31:0]

0xx

0xx

0x0

0x20

0xx

0x00000000

0x2

0x00000000

1

0x13

x

1

0

0x0

x

1

0x00

0x00

0x05

0x00000000

0x00000000

0xx

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

00

50.00

100.00

150.00

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main*

86

6

AWS

Port : 5500

[illegible]

[illegible]

[illegible]

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VCD

Zoom

Cursor

HOME

<

>

END

Filter tree

TOP

TOP.audio_soc_top.u_core.wb_data [31:0]

TOP.audio_soc_top.u_core.reg_we

TOP.audio_soc_top.u_core.alu_s

TOP.audio_soc_top.u_core.mem_reg

TOP.audio_soc_top.u_core.alu_op [1:0]

TOP.audio_soc_top.u_core.alu_ctrl [3:0]

TOP.audio_soc_top.u_core.funct3 [2:0]

TOP.audio_soc_top.u_core.funct7 [6:0]

TOP.audio_soc_top.u_core.u_alu.a [31:0]

TOP.audio_soc_top.u_core.u_alu.b [31:0]

TOP.audio_soc_top.u_core.u_alu.alu_ctrl [3:0]

TOP.audio_soc_top.u_core.u_alu.result [31:0]

TOP.audio_soc_top.u_core.u_alu.zero

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TOP.audio_soc_top.u_core.u_ctrl.reg_write

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TOP.audio_soc_top.u_core.u_ctrl.mem_to_reg

TOP.audio_soc_top.u_core.u_ctrl.alu_op [1:0]

TOP.audio_soc_top.u_core.u_regfile.clk

TOP.audio_soc_top.u_core.u_regfile.we

TOP.audio_soc_top.u_core.u_regfile.rs1 [4:0]

TOP.audio_soc_top.u_core.u_regfile.rs2 [4:0]

TOP.audio_soc_top.u_core.u_regfile.rd [4:0]

TOP.audio_soc_top.u_core.u_regfile.wd [31:0]

TOP.audio_soc_top.u_core.u_regfile.rd1 [31:0]

TOP.audio_soc_top.u_core.u_regfile.rd2 [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[1] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[2] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[3] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[4] [31:0]

TOP.audio_soc_top.u_core.u_regfile.rf[5] [31:0]

0xx

x

x

x

0xx

0xx

0x0

0x20

0xx

0x00000000

0x2

0x00000000

1

0x13

x

1

0

0x0

x

1

0x00

0x00

0x05

0x00000000

0x00000000

0xx

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0xx

0xx

0xx

0x0

0x00

0x00000000

0x2

0x00000000

0x00

0x0

0x00

0x00

0x00

0x00000000

0x00000000

0xx

0x00000000

0x00000000

0x00000000

0x00000000

00

500.00

1000.00

1500.00

2000.00

2500.00

3000.00

3500.00

4000.00

4500.00

5000.00

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main* 86 6 AWS

Port : 5500